

An Advanced Spectroscopy Lab That Integrates Art, Commerce, and Science as Students Determine the Electronic Structure of the Common Pigment Carminic Acid

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S Supporting Information

ABSTRACT: Carminic acid and its metal derivatives have been used widely as pigments for fabrics and art, and more recently as a colorant for food. An undergraduate teaching laboratory is described in which students are instructed to design and execute experimental studies to obtain detailed information about the electronic structure, metal complex formation, redox properties, and photochemical stability of extracted carminic acid. The lab maintains room for student innovation and is well-suited to upper-level undergraduates in an advanced spectroscopy lab.

Students are invited to apply knowledge previously gained through highly directed experiments to the analysis of an unfamiliar, complex, and relevant problem. The laboratory lends itself to flexibility in implementation but is designed to deepen students' understanding of UV-vis spectroscopy, fluorescence spectroscopy, IR spectroscopy, electrochemistry, pH and metal spectrophotometric titrations, and experimental determination of the kinetic behavior of UV-induced decomposition.

KEYWORDS: Upper-Division Undergraduate, Interdisciplinary/Multidisciplinary, Inquiry-Based/Discovery Learning, Electrochemistry, Fluorescence Spectroscopy, Spectroscopy



INTRODUCTION

Carminic acid (Figure 1) is a natural compound extracted from cochineal insects.¹ The chromophore in this molecule is the conjugated π system extending across the central anthraquinone ring system.

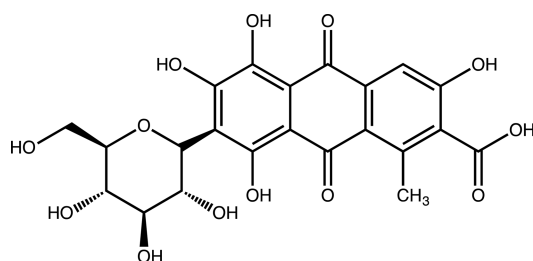


Figure 1. Carminic acid.

Carminic acid can combine with various metals to form the pigment carmine.² In industry, it is commonly complexed with aluminum to produce a purple/pink precipitate (with calcium as a counterion). Carmine's color differs depending on the metal it is complexed to. Having a number of potential tautomers adds to its structural complexity.³

For centuries, carminic acid and carmine have been used as dyes. The source material is cochineal, a blood-like fluid found within the cochineal insect.^{2b} These insects live on various species of cacti and are harvested directly from the plants they inhabit. The dye was developed in southern and central

America and exploited by the Spanish after the conquest of these regions.^{2b} (Note: There is an old-world version of carmine, from a scale insect native to the Mediterranean region. The insect is called kermese, and the acid is kermesic acid. But the color, which is a bit different, is still called carmine. Titian used it.)

Currently, carminic acid and carmine have a multitude of applications in the modern world. They are used as nontoxic food additives and biological stains and are finding new uses as electrochemical modifiers and photosensitizers. Several years ago, a public outcry caused Starbucks to eliminate cochineal-based colorants from its beverages.⁴ Carmine and carminic acid have been characterized by a variety of techniques, primarily to help with the identification of art work or food samples.⁵ Collection, distribution, and production methods vary, and variability in carmine production remains a concern.^{2b}

For this laboratory project, students were first introduced to some of the science and applications of carmine, and then asked to research methods to produce carminic acid and to develop a plan to use one or more of the spectroscopic techniques they had previously studied to characterize an aspect of the electronic structure of carminic acid or carmine. Students were given as much cochineal-containing lice as they wanted and 3–4 weeks in which to complete their work.

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This laboratory exercise was the culmination of a semester-long advanced spectroscopy lab course and was developed to provide students with the opportunity to apply knowledge they had learned in highly structured analyses to an open-ended problem with significant relevance to art and commerce. Before shifting to this project, students spent about two-thirds of the semester following well-established protocols to make a series of standard measurements using a range of spectroscopic techniques.

Similar spectroscopy-based laboratories undertaken elsewhere have received positive feedback from students.⁶ In one example, X-ray fluorescence, infrared spectroscopy, and UV–vis absorption spectroscopy were used to identify forged art.^{6a} Both art history students and chemistry students benefitted; art history students were able to understand how chemistry knowledge can be applied to paintings, and chemistry students learned how to utilize instruments to judge a painting's authenticity.^{6a} Another article published in this *Journal* described the use of fluorescence and Raman spectroscopy to identify the time frame in which two paintings were made, allowing their authenticity to be determined.^{6c} Raman spectroscopy and surface-enhanced Raman scattering have also been used in laboratory exercises focused on identifying colorants in art.^{6d} Another recent article in this *Journal* highlights the utility of food dyes as a way of connecting core chemistry concepts to real examples that are of interest to students.⁷

All of these papers provide evidence that problem-based laboratory experiences strengthen problem-solving and reasoning skills.⁸ This approach has the additional advantage that it allows students to apply techniques they learned previously to real-world problems and that, in turn, enables them to understand the relevance of the skills they are learning. For example, students in our course had already studied the fluorescent properties of $[\text{Ru}(\text{bpy})_3]^{2+}$ using detailed instructions provided in our lab manual. When it came time to characterize carminic acid using fluorescence spectroscopy, the rationale behind excitation and emission wavelength selection was unclear to many students. Revisiting fluorescence spectroscopy in this new context encouraged students to think more deeply about the core concepts underlying the exploration of excited optical states of chromophores, and draw conclusions.

In this paper, we summarize the experiments undertaken and suggest related experiments that might be of interest. All of the experiments reported here were repeated at least three times by different groups of students. All experimental results were confirmed by a faculty member using authentic carminic acid. Extensive experimental details and complete characterization information are provided in the [Supporting Information](#).

RESULTS AND DISCUSSION

Here, we provide a summary of our results and a discussion of their significance. Detailed information, including a student handout, copies of actual spectra and voltammetric scans, and reports of student assessment, is provided in the [Supporting Information](#).

Extraction Methods

The method by which carminic acid is extracted can impact the chemical composition of the material released. We used three extraction methods, each yielding material with different optical properties. We found that carminic acid produced by extracting

the ground cochineal with water yielded material that most closely matched the commercially available standard.

Spectroscopic Characterization Methods

UV–Vis Spectroscopy. Carmine and carminic acid can be characterized by UV–vis spectroscopy.^{9,10} The characteristic peaks [with molar absorptivities ($\text{M}^{-1} \text{cm}^{-1}$)] for absorption spectrum of carminic acid in water are 531 (1700), 496 (2700), 468 (2600), 316 (4100), 279 (12,200), 222 (7200). The characteristic peaks [with molar absorptivities ($\text{M}^{-1} \text{cm}^{-1}$)] for absorption spectrum of carmine are 554 (4400), 518 (5800), 480 (4400), 334 (8200), 281 (17,900), 226 (19,500).

The absorption spectra of carminic acid and carmine are pH-dependent and broad; in general, the longest wavelength peaks shift upon metal complexation.¹¹ The lowest energy transition is a π to π^* transition with some charge transfer (CT) from the phenolic donor to the quinoid acceptor group.¹⁰ Metal complexation has relatively little effect on the electronic spectrum although, as described in more detail below, it impacts the charge transfer transition coupled to the π to π^* transition and so red-shifts the absorbance.¹² Even small shifts in the visible spectrum lead to dramatic changes in perceived color.

pH Titrations Followed by UV–Vis Spectroscopy.

Because it has multiple acidic protons, carminic acid has a pH-dependent absorption profile that lends itself to careful characterization by UV–vis/pH titrations.¹³ We note an isosbestic point at 503.9 nm in the titration between pH 2 and 8.

Fluorescence Spectroscopy. Carminic acid is a fluorophore that emits light from an excited singlet to a single ground state.^{14,15} Fluorescence analysis is a common method for nondestructively analyzing art work and has been used to identify carmine-based pigments in artifacts.¹⁶ Many of the carminic acid/metal complexes also luminesce, generally with excitation and emission wavelengths red-shifted relative to those of the free base.¹⁷ The complex can undergo an excited state proton transfer leading to multiple peaks in the fluorescence spectrum.^{10,17,18}

ATR-FTIR Spectroscopy. Peaks in the C=O stretching region ($1650\text{--}1800 \text{ cm}^{-1}$) are observed for both carminic acid and carmine.

Photoirradiation of Carminic Acid

Carmine and carminic acid have been considered less than ideal pigments for paintings due to their photochemical instability (although they appear in some very famous paintings by great masters).¹⁹ However, they are thermally and photochemically stable enough to be used widely as food and beverage additives.¹⁰ Photochemical reactions are pedagogically useful because they introduce students to a host of concepts that they might not otherwise encounter such as actinometry and quantum yields. The photochemical bleaching of carminic acid is easy to follow. We used UV–vis absorption spectroscopy to study its stability, but fluorescence spectroscopy is also sensitive to changes in electronic structure.²⁰ We irradiated a carminic acid sample with a UV lamp and compared the absorbance of the irradiated sample with a dark (shielded from all light) and light (exposed to ambient room light but not the UV lamp) control. We did not explore the photochemical stability of carminic acid as a function of pH, but we noticed that high pH samples were unstable even when protected from light; hence, a base-catalyzed ring-degradation process must exist. The photodecomposition is thought to result from a ring

opening of the anthraquinone moiety.¹⁹ Oxygen does not appear to play a role in the photochemical decomposition, as deaeration with argon does not change the photodecomposition rate.¹⁰ We monitored the absorbance change at 495 nm and observed a first-order decomposition reaction.

Cyclic Voltammetry Experiments

Carminic acid has interesting electrochemical properties that are suited well to the introduction of cyclic voltammetry into the undergraduate curriculum. The hydroxyanthraquinone core can participate in multielectron redox processes.²¹ The potentials corresponding to the reduction and oxidation of the quinone and catechol moieties are highly dependent on the protonation state of the ligand.

We limited our electrochemical measurements to the oxidative window of +0.3 to 0.85 V (vs Ag/AgCl) at glassy carbon electrodes. A single quasireversible redox couple ($E_{\text{mid}} = 0.56$ V) was observed under acidic conditions (pH 2). Peak currents diminished slightly with consecutive cycling, indicative of surface adsorption. The influences of concentration, scan rate, and pH on the voltammetric response were studied.

Determination of Stoichiometry of Y^{3+} –Carminic Acid Complex Using Job's Method

The coordination chemistry of carminic acid is an underexplored area. We carried out a model experiment titrating carminic acid with a solution of yttrium and monitored complex formation by a (dramatic) change in the visible spectrum. The resulting data were fitted successfully using the method of Job. We present data for only one metal ion, but the same approach could be used for other ions. There are literature reports on studies of the stoichiometry of binding of scandium,²² gallium,²³ and lanthanum.²⁴

HAZARDS

These experiments present few hazards to the student. Students should use appropriate PPE (personal protective equipment) at all times, which includes wearing lab coats, goggles, and gloves. Appropriate laboratory attire is expected, which include clothes that leave no exposed skin, closed toed shoes, securing long hair, and wearing no dangling jewelry or scarves. Acids and bases should be handled appropriately. A hot plate is used to extract the carminic acid and students should use it with care. The UV lamp should be used with safety goggles rated for UV light and students should not look directly at the lamp. Students should read instructions and be trained on all instruments before operating.

CONCLUSIONS

This set of proposed activities for an advanced spectroscopy lab provides opportunities for students to design a small spectroscopic or analytical project within the context of carminic acid chemistry. The interrelated nature of the proposed experiments enables teams of students to identify specific experiments they want to perform while the class as a whole shares references and data to arrive at an integrated understanding of the electronic structure and chemical behavior of carminic acid. In our lab, the historical and cultural uses of cochineal-based products captured the interest of everyone who participated. (We direct the reader's attention to some videos that highlight the history and native extraction processes for creating carminic acid.²⁵) The variety of hues produced during the course of the three week period brought joy to everyone

who walked past the laboratory or saw the students' Instagram accounts (see Figure 2).



Figure 2. Solutions of carminic acid at pH 1–14.

The project could be expanded in a number of directions, depending on institutional resources and instructor expertise. We point out a few of these below. Published procedures exist for other extraction methods including some that use supercritical fluids, which we were not in a position to try but which might be appropriate in other laboratory settings.²⁶ There are also efforts to reproduce older extraction methods to recreate a historical context.²⁷ Some references suggest that various types of surface-enhanced Raman spectroscopy might be useful for probing the electronic structure of carminic acid and carmine.²⁸ HPLC–MS has been used to characterize organic dyes and can be employed to separate carminic acid from related compounds.^{29,30} There is a wealth of coordination chemistry with different metal ions that could be explored, and the delight of seeing different colored complexes form would be rewarding.^{23,31} Excited state chemistry could be explored in more detail,^{18,32} as could computational chemistry of carminic acid.¹² Carminic acid can also be adsorbed to solid supports to serve as a photosensitizer, opening up the exploration of photosensitized solar cells³³ and further electrochemical characterization.

In summary, this activity offers advanced undergraduate students the opportunity to design and implement a spectroscopic or analytical project using a commercially and culturally significant chromophore.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available on the ACS Publications website at DOI: 10.1021/acs.jchemed.6b00644.

Data provided includes UV–vis spectra, FTIR spectra, fluorescence spectra, Job's method data, UV decomposition data, cyclic voltammetric scans and accompanying data analysis, student handout, and reports of student assessment (PDF, DOCX)

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Notes

The authors declare no competing financial interest.

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Carminic acid solutions photographs were taken by Jim Austin.

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